

Update - Differential Analysis of LLMs in Text2SQL Generation

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Abstract

Large Language Models (LLMs) have shown remarkable capabilities in translating natural language questions into executable SQL queries, a task known as Text2SQL. In this work, we conduct a systematic evaluation of several state-of-the-art LLMs and two distinct Text2SQL frameworks: agentless and agent-based pipelines. Our experiments assess accuracy, consistency, and execution robustness on the BIRD-SQL Dev Set. By enabling users to query structured databases through natural language, effective Text2SQL systems have the potential to democratize data access, allowing non-developers to retrieve complex information from structured databases without needing specialized technical skills. Our study aims to inform future design choices for building more reliable and accessible natural language interfaces to databases.

1 Introduction

As large language models (LLMs) continue to evolve and improve, an interesting application of LLMs is in the generation of SQL queries from natural language. This is known as Text2SQL. Text2SQL can enable non-experts to extract valuable information from a database without having to learn complex query languages like SQL. By simply describing the data they want in plain English, users can leverage LLMs to translate their requests into structured queries that interact directly with a database. This has the potential to democratize data access across organizations, reduce reliance on technical teams for routine data retrieval tasks, and improve overall efficiency in data-driven decision-making.

In this project, we compare the performance of different LLMs in generating SQL queries from natural language. We explore 2 different frameworks, Agentless and Agent-based, and 3 different LLMs, Claude 3.5 Sonnet, GPT-4o mini, and Llama3.1 8B. In the Agentless framework, the descriptions of all tables in the database (relevant and irrelevant) are passed as context to the LLM when generating the SQL query given the natural language query. In the

Agent-based framework, there are 3 agents that handle specific tasks such as determining the number of relevant tables, retrieving the descriptions of the relevant tables, and generating a SQL query given a natural language query. We run our Agentless and Agent-based frameworks with the 3 different LLMs on the BIRD-SQL Dev Set benchmark.

By systematically analyzing various LLMs and frameworks, we aim to understand how model choice and pipeline design impact the accuracy, consistency, and execution success of generated SQL queries. Our findings can help guide the development of more reliable and accessible Text2SQL systems, ultimately making database interaction easier and more intuitive for a broader range of users.

2 Previous Work

Text2SQL has been an active area of research, with approaches evolving alongside advances in natural language processing. Early methods primarily relied on rule-based systems [?] and semantic parsing [?], which required significant domain-specific engineering and struggled to generalize across different database schemas. With the rise of deep learning, encoder-decoder architectures and pre-trained language models began to dominate, enabling more flexible mappings from natural language to SQL queries.

Recent work has explored building Text2SQL pipelines that leverage LLMs to generate SQL queries through structured multi-step processes. CHASE-SQL [?] and XiYan-SQL [?] both implement multi-agent frameworks in which one agent retrieves relevant schema information, another generates candidate queries, and a final agent selects the most appropriate SQL statement. CHASE-SQL employs methods such as divide-and-conquer chain-of-thought (CoT) reasoning, query-plan-based CoT reasoning, and on-line synthetic example generation to enhance candidate generation. In contrast, XiYan-SQL integrates supervised fine-tuning of multiple specialized generators alongside in-context learning (ICL) with skeleton-based example selection to produce high-quality and diverse SQL candidates. CHASE-SQL and XiYan-SQL rank 2nd and 4th respectively for the BIRD benchmark.

While these recent frameworks demonstrate the effectiveness of multi-agent pipelines for improving Text2SQL performance, there remains value in systematically comparing different LLMs and framework designs under a common experimental setting. In this work, we conduct an empirical evaluation of both agentless and agent-based pipelines across multiple LLMs, aiming to better understand how model choice and pipeline structure influence SQL generation quality on the BIRD-SQL Dev Set. Specifically, we evaluate not only a larger model such as Claude 3.5 Sonnet, but also smaller, potentially more economical models like GPT-4o Mini and Llama 3.1 8B. To keep the evaluation feasible, we employ a simplified version of the agent-based framework, utilizing a single generator rather than multiple specialized generators. Additionally, to assess the consistency of each model and framework combination, we run each configuration three times.

3 Methodology

In this section, we outline the methodology used for our differential analysis of LLMs in Text2SQL generation.

3.1 Model Selection

For this study, we selected the three Large Language Models (LLMs) Claude 3.5 Sonnet, GPT-4o mini, and Llama3.1 8B. These models for their widespread adoption and diversity in model size. Particularly, Claude 3.5 Sonnet has 175 billion parameters; whereas, GPT-4o mini and Llama3.1 8B have 8 billion parameters [?]. Notably, Llama3.1 8B is open-source, while GPT-4o Mini and Claude 3.5 Sonnet are closed-source. By comparing these models, we aim to evaluate how performance varies not only with model size but also between open-source and closed-source LLMs.

Temperature controls the degree of randomness in the text generated by large language models during inference (<https://www.ibm.com/think/topics/llm-temperature>). The default temperature for Claude 3.5 Sonnet is 1.0, which represents the maximum level of randomness. For GPT-4o mini, the default temperature is 0.7; however, in this study, we increase it to 1.0 to encourage greater diversity in the SQL queries generated across three runs, thereby improving the chances of obtaining a correct result. For Llama 3.1 8B, we retain the default lower temperature, as smaller models are generally more prone to hallucinations when exposed to higher levels of randomness.

3.2 Framework Selection

There are many ways that Text2SQL can be implemented. In this study, we compare the performance differ-

ence between 2 frameworks, one that performs without AI agents (Agentless) and one that utilizes a series of AI agents (Agent-based).

1. **Agentless:** In this framework, an LLM to generate the response directly. This is synonymous with the use of a chatbot. For this framework, we will provide the descriptions of every table in the database and the natural language query to the **LLM**. Then we will run the generated SQL query and evaluate the results based on accuracy, execution time, and resource utilization. The Agentless pipeline can be seen in 1
2. **Agent-based:** In this framework, there are distinct agents that each play a different role in the generation of the response. There are 3 agents: **Relevance Agent**, **Retrieval Agent**, and **Query Agent**. The **Relevance Agent** is responsible for determining how many tables are relevant to a natural language query. The **Schema Agent** will be responsible for retrieving the descriptions of the relevant tables from the database given the natural language query. The **Query Agent** will be responsible for generating the SQL query given the relevant tables' descriptions and the natural language query. Then we will run the generated SQL query and evaluate the results based on the same criteria as the Agentless framework. The agent-based pipeline can be seen in 2

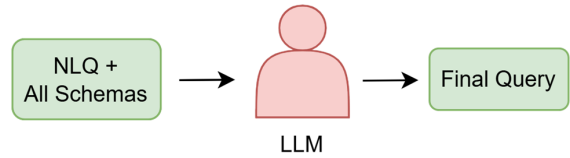


Figure 1: Agentless Text2SQL Pipeline

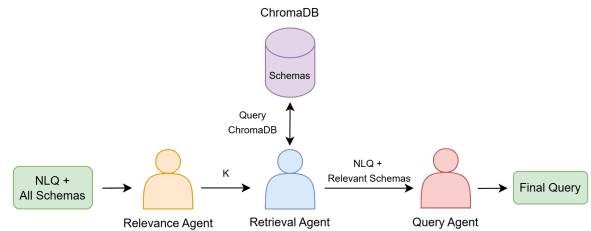


Figure 2: Agent-based Text2SQL Pipeline

The main difference between Agentless and Agent-based framework is that during the Text2SQL generation, the descriptions of all tables are used for Agentless, whereas only

the descriptions of relevant tables are used for Agent-based. The comparison between these frameworks can provide insight into the trade-offs between simplicity and modularity, particularly in terms of scalability and accuracy – considering Agentless make less calls to the LLM than Agent-based. Specifically, we aim to explore whether introducing specialized agents leads to more accurate query generation, or if the added complexity results in diminishing returns compared to a simpler, monolithic approach. This evaluation can help guide future development of LLM-powered database interfaces.

3.3 Benchmark Dataset Selection

For this study, we use the BIRD-SQL Dev Set as the benchmark for evaluation. BIRD-SQL is a widely adopted benchmark in the Text2SQL community, known for its diverse database schemas, realistic query complexity, and standardized difficulty categorizations (simple, moderate, and challenging). Its acceptance as a standard benchmark enables fair comparisons across different methods and models.

The BIRD-SQL Dev Set includes table descriptions for every database, a collection of natural language questions (NLQs), their corresponding ground-truth SQL queries, and assigned difficulty levels. For our experiments, we focus on the "student-club" database and sample 30 questions, comprising 18 simple, 9 moderate, and 3 challenging examples. This curated subset serves as the ground truth against which we evaluate the performance of the selected LLMs and frameworks.

3.4 Evaluation Metrics

We evaluate the performance of each Model+Framework combination across three main metrics: **correctness**, **consistency**, and **execution error rate**.

- **Correctness** measures how often the generated SQL query produces the correct result for a given natural language question (NLQ). To assess accuracy, we execute the generated SQL query and the corresponding ground-truth SQL query from BIRD-SQL on the database, and determine correctness by comparing their execution results. A generated query is considered correct if its execution result exactly matches the execution result of the ground-truth query. We categorize accuracy based on the number of successful generations out of three attempts for each query: 0/3, 1/3, 2/3, or 3/3 correct generations.
- **Consistency** captures the stability of a model's outputs across multiple runs. A Model+Framework combination that achieves 3/3 correct generations for a query is

considered highly consistent, while fluctuating results (e.g., 1/3 or 2/3 correct) indicate lower consistency.

- **Execution Error Rate** is the percentage of generated SQL queries that fail to execute, typically due to syntax errors, invalid table references, or improper SQL structure. A lower execution error rate reflects greater robustness and syntactic reliability. A higher execution error rate may indicate hallucinations, misunderstandings of the database schema, or general instability in query generation.

Additionally, we compare the overall performance between the agentless and agent-based frameworks to explore the trade-offs between simplicity (agentless) and modular specialization (agent-based). Each Model+Framework combination is tested on the 30 sampled queries, with three independent runs per query to capture variability.

4 Results

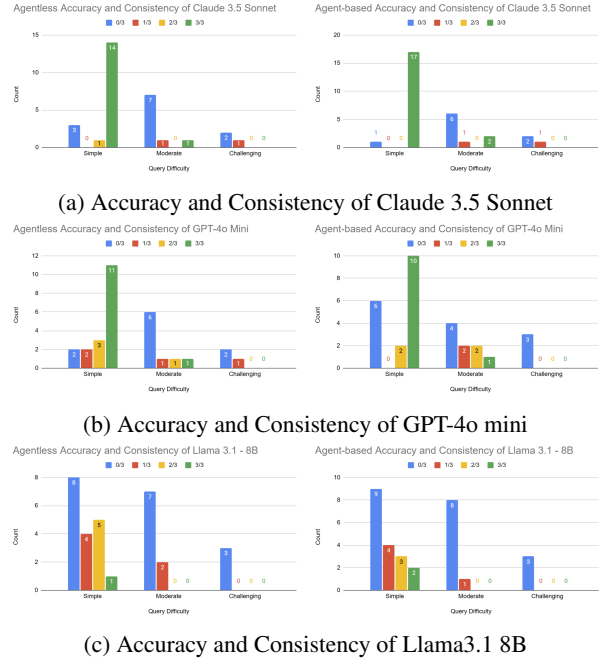


Figure 3: Accuracy and Consistency

5 Conclusion

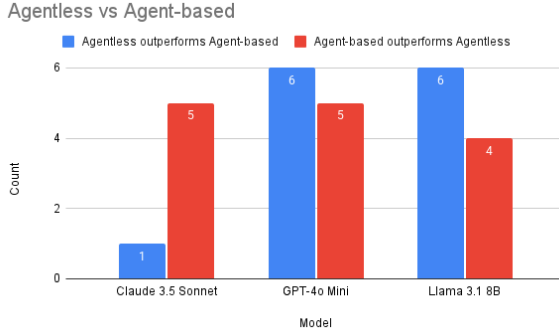


Figure 4: Agentless vs Agent-based Comparison

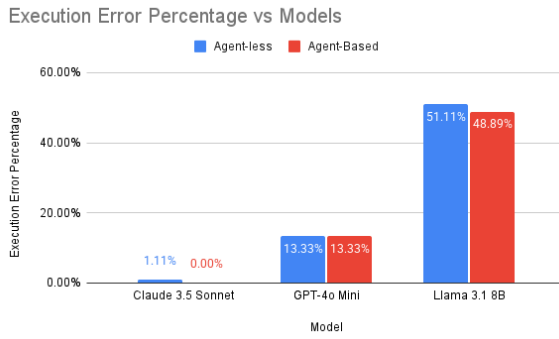


Figure 5: Execution Error Comparison

6 Additional Notes

We will continue running this study on Llama 3 8B and another LLM, which is still to be determined. As mentioned earlier, we also plan to omit the Chinook benchmark moving forward and expand the size of the BIRD-SQL benchmark. This change provides several benefits. First, manually creating natural language queries and their corresponding ground truth answers is prone to errors, especially given our limited SQL expertise. Second, BIRD-SQL offers natural language queries paired with corresponding SQL queries, enabling us to scale our system to a larger benchmark since we no longer have to manually determine the ground truth answers. Finally, BIRD is a widely used benchmark in Text-to-SQL research and its diversity in schema and query complexity also allows for more robust evaluation of framework and model performance.